

Open-Source SNN and WBE Projects

Kurzantwort: Ich habe die wichtigsten Open-Source-Projekte und aktuelle wissenschaftliche Übersichten zu gehirn-inspirierten Architekturen, Spiking Neural Networks (SNN) und Whole-Brain-Emulation zusammengetragen; unten findest du eine vergleichende Übersicht, konkrete Implementierungsansätze für Isaac und mehrere neuartige Ideen, die noch wenig verfolgt sind.

Vergleich: relevante Open-Source Projekte (Kernattribute)

Projekt	Fokus	Reife	Codebasis / Sprache	Best Use for Isaac
BrainCog / Brain-Cog	Multi-scale SNN cognitive engine	Aktiv, forschungsorientiert	Python (SNN libs)	Cognitive modules, episodic memory emulation.[^1]
Spiking frameworks (SpikingJelly, snnTorch, Norse)	SNN training & simulation	Mature libraries, benchmarks exist	PyTorch / JAX	Low-power event processing, temporal coding.[^2][^3]
Open Neuromorphic (community)	Ecosystem, hardware /software bridge	Active community & resources	Links zu SNN tools & hardware	Roadmap, benchmarks, deployment guidance.[^4]
Carboncopies / BrainGenix (WBE tools)	Whole-brain emulation tooling & evaluation	Research platform, modular stacks	C++ / Python mixed	Long-term emulation experiments, validation metrics.[^5]

Aus deinem System-Analyse-Dokument: "Isaac is a personal AI core system designed for privacy-first, trust-based autonomous operation." "The system works well in production but shows architectural strain:"

Konkrete Integrations- und Implementierungsvorschläge

1) Hybrid SNN + ANN memory layer (BLAU)

- **Warum:** SNNs kodieren Zeit/latency nativ; kombinieren sie mit ANN-Vektor-Embeddings für semantic recall reduziert Redundanz und passt zu Isaac's privacy-first lokalem Speicher.

Implementierung: nutze snnTorch/SpikingJelly für temporal encoders; synchronisiere mit existing vector_memory via adapter.^{[2][3]}

2) Neuromorphic event loop für GRÜN (Execution)

- **Warum:** Event-driven Ausführung (sparse spikes) senkt Latenz und Energie; passt zu BackgroundLoop/Watchdog. **Implementierung:** mappe kernel events auf SNN spikes; teste mit Rockpool/Lava-DL backends^{[2][4]}.

3) Modulare ROT Orchestrator inspiriert von cortical microcircuits

- **Warum:** Reduziert Monolith (isaac_core heavy coupling) durch biologisch inspirierte, hierarchische control-modules (local inhibitory/excitatory motifs) - bessere fault isolation. *Phase-wise extraction as in your plan* (document).

Neuartige, wenig erforschte Ansätze (ausführlich)

1. **Dendritic-style local planners:** implementiere per-module multi-compartment neuron models (dendritic gating) als lightweight policy networks - erlaubt context-dependent gating without large symbolic controllers. (Inference: could improve compositional planning; needs custom neuron models and sparse training.^[6])
2. **Event-based RAG (retrieval-augmented generation) using spike timing:** encode retrieval cues as spike patterns; retrieval matches via temporal cross-correlation rather than vector cosine - promising for privacy/local matching and robustness to drift. (Novel; requires research prototype.)
3. **Functional emulation sandbox (mini-WBE modules):** small-scale BrainGenix-style modules to validate functional continuity of Isaac subsystems (memory, planning) before full integratio^[5]n.

Risiken & nächste Schritte

- **Risiken:** SNN training complexity; hardware mismatch; evaluation/validation gaps (WBE ethics & metrics^{[2][5]}).
- **Empfohlen:** 1) Proof-of-concept Hybrid SNN memory (4-8 weeks), 2) integrate event loop in isolated GRÜN testbed, 3) use Open Neuromorphic community for benchmark^{[4][1]}s.

Wenn du willst, setze ich **PoC-Scope** und eine priorisierte Task-Liste für Phase-2-Integration (inkl. test-cases und metrics) direkt um.

References (6)

^[1]: brain-inspired-ai · GitHub Topics · GitHub. <https://github.com/topics/brain-inspired-ai>

^[2]: Spiking Neural Network (SNN) Frameworks - Open Neuromorphic. <https://open-neuromorphic.org/neuromorphic-computing/software/snn-frameworks/>

[^3]: *GitHub - fangwei123456/spikingjelly: SpikingJelly is an open-source*

<https://github.com/fangwei123456/spikingjelly>

[^4]: *Neuromorphic Computing - Open Neuromorphic.* <https://neural-loop.github.io/open-neuromorphic.github.io/neuromorphic-computing/>

[^5]: *GitHub - carboncopies/BrainGenix-NES: [Mirror of https://gitlab*

<https://github.com/carboncopies/BrainGenix-NES>

[^6]: *Brain-inspired Artificial Intelligence: A Comprehensive Review.*

<https://arxiv.org/html/2408.14811v1>